reasoning

artificial intelligence · prof. dr. nicolas meseth

1 thinking is writing
2 from a trick to a skill
3 learning from rewards
4 longer or more often
5 what it is not
6 why agents need it

some questions
are easier
on paper than
in your head.



image: ai-generated (gpt-image-2.5-flare)

part 1

thinking is writing

a tractor and a trailer cost 11,000 € together, and the tractor costs 10,000 € more than the trailer. what does the trailer cost?

500 €

1,000 €

1,500 €

10,000 €

live.hsos.uber.space/q

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live.hsos.uber.space/q

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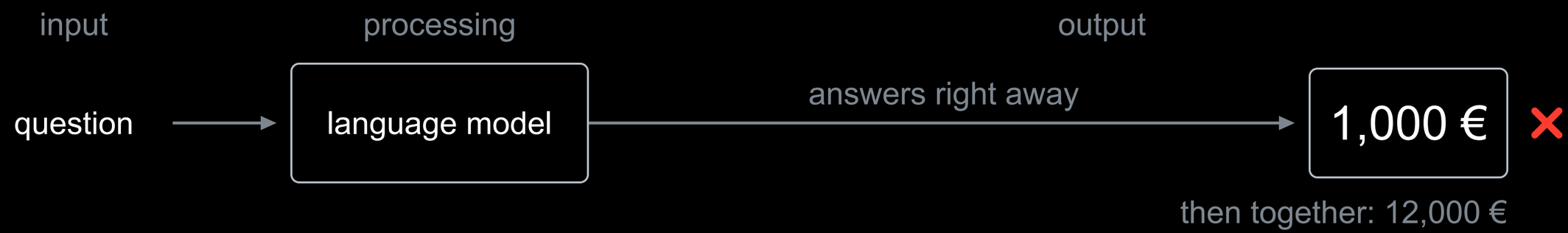
1,000 €

1,500 €

10,000 €

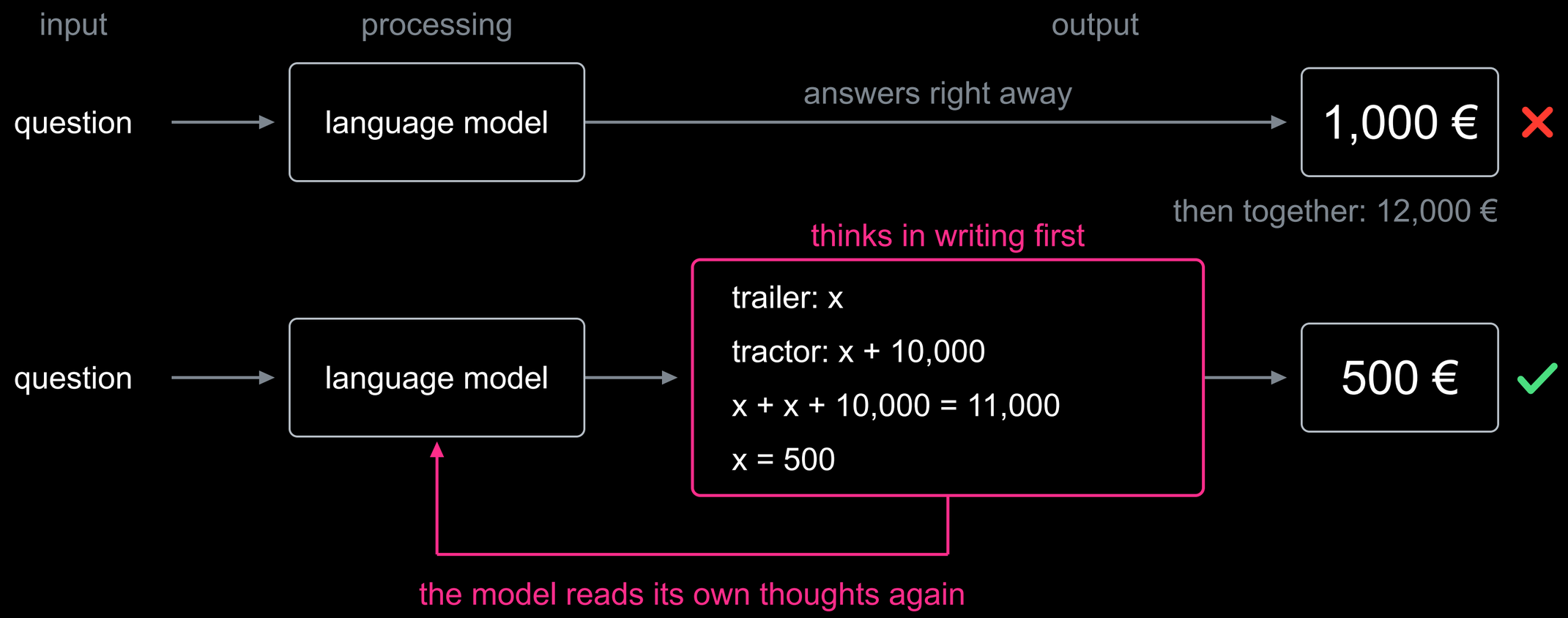
live.hsos.uber.space/q

answer at once, or think first



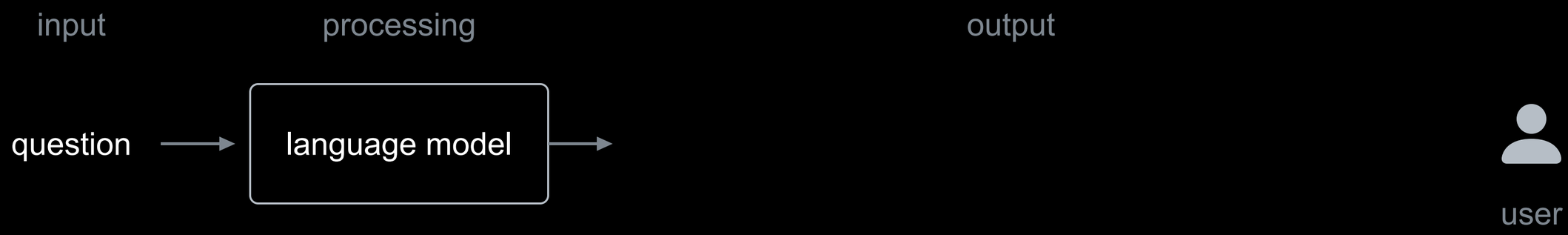
then together: 12,000 €

answer at once, or think first



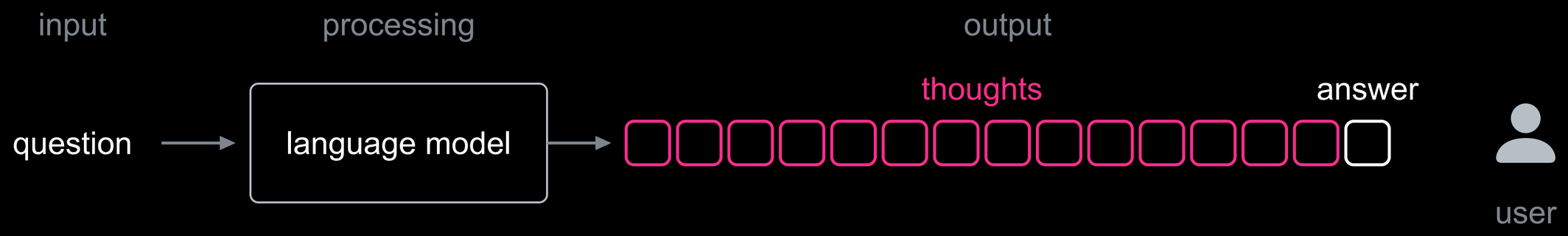
a reasoning model is the same transformer predicting the same tokens, and what changed is what it has learned to write before the answer.

more tokens, more computation



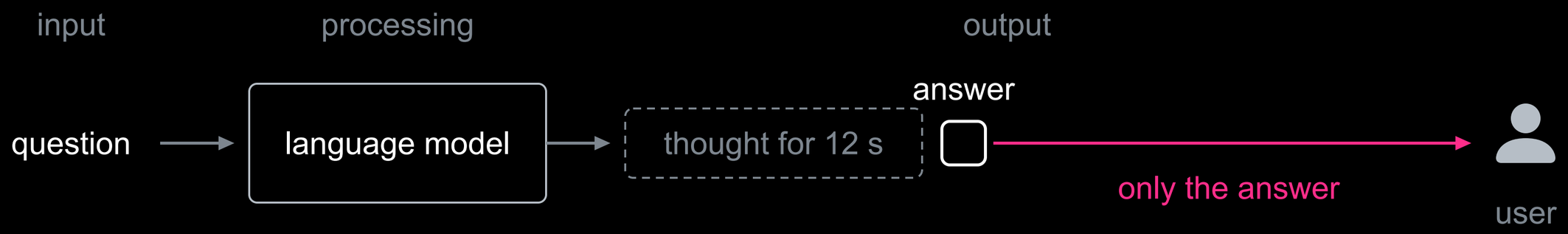
every token is one pass through the model, so 500 tokens of thought are 500 passes

more tokens, more computation



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more tokens, more computation



the app folds the thoughts away or summarises them, and the user sees only the answer

context

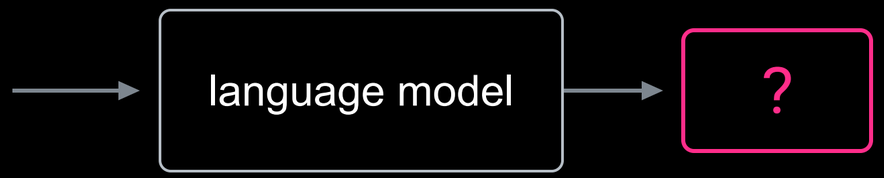
```
tractor + trailer = 11,000 €
tractor = trailer + 10,000 €

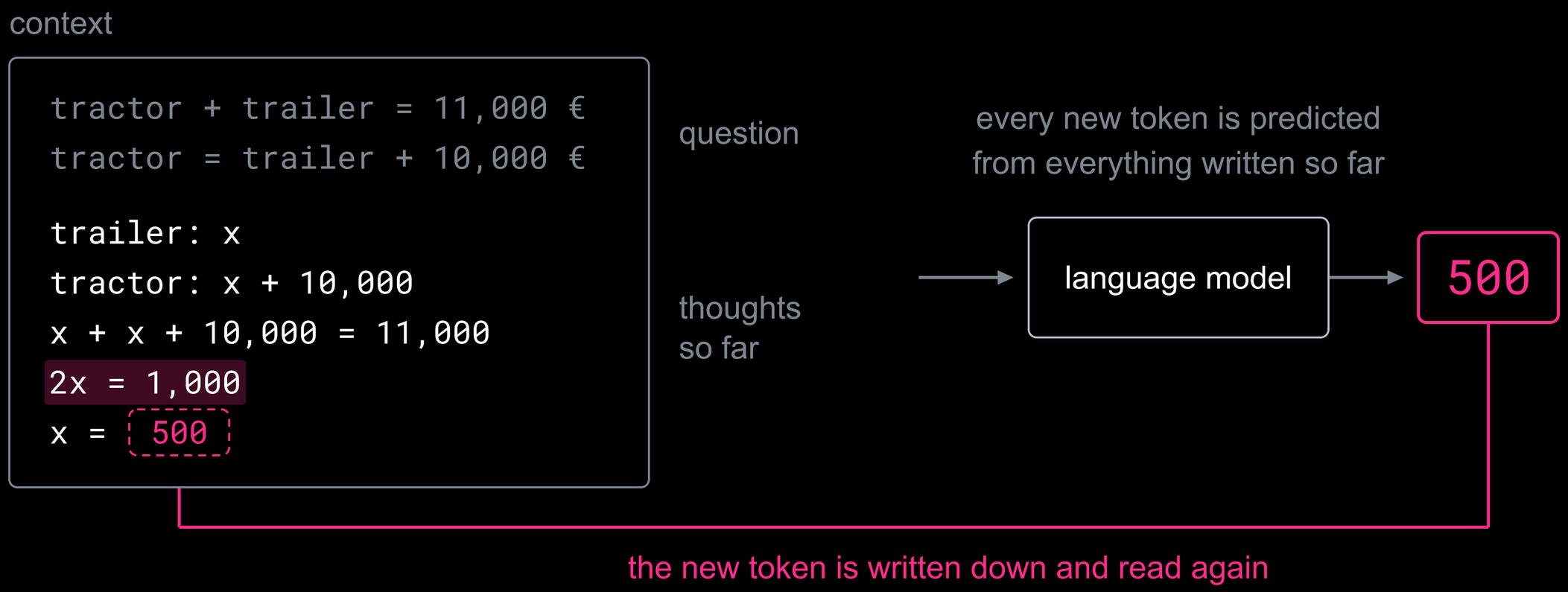
trailer: x
tractor: x + 10,000
x + x + 10,000 = 11,000
2x = 1,000
x =
```

question

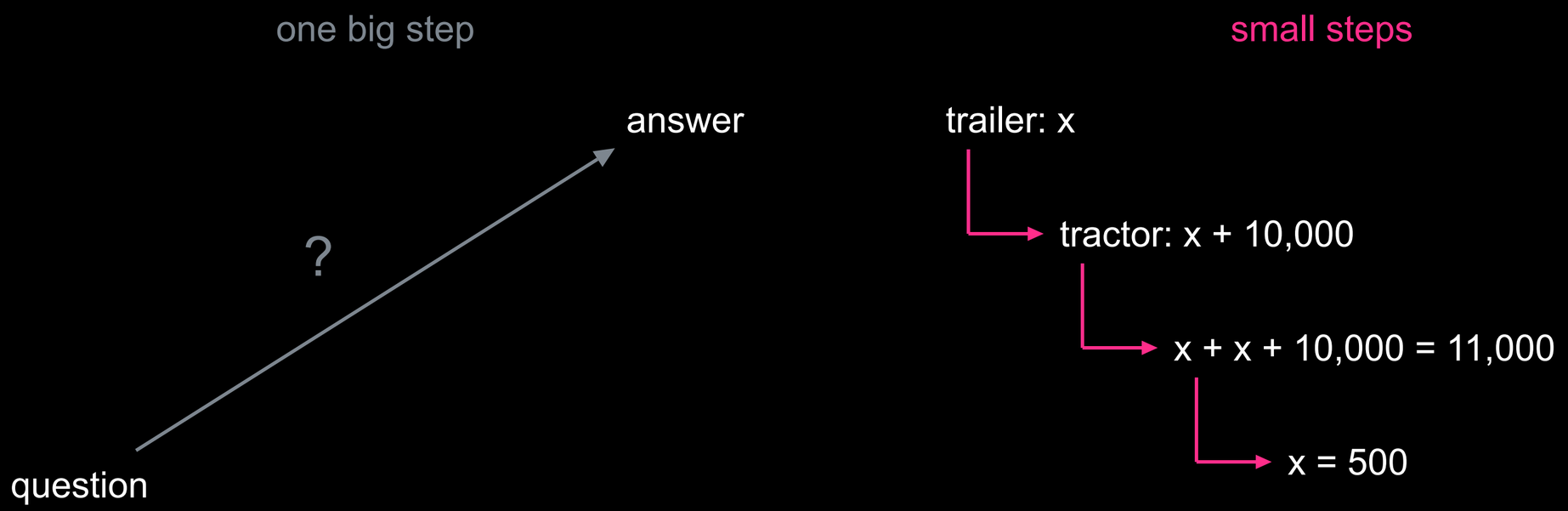
thoughts so far

every new token is predicted from everything written so far




context
tractor + trailer = 11,000 €
tractor = trailer + 10,000 €
trailer: x
tractor: x + 10,000
x + x + 10,000 = 11,000
2x = 1,000
x = 500
question
thoughts so far
every new token is predicted from everything written so far
language model
500
the new token is written down and read again

small steps are easier than one big step



```
trailer: 1,000
tractor: 11,000
together: 12,000
```

```
trailer: 1,000
tractor: 11,000
together: 12,000

wait, that is 12,000, not 11,000.
trailer: x, tractor: x + 10,000
```

part 2

from a trick to a skill

few-shot: solved examples in the prompt

prompt

```
Q: Roger has 5 tennis balls. He buys 2 cans of 3 tennis balls each.
   How many tennis balls does he have now?
A: The answer is 11.

Q: The cafeteria had 23 apples. They used 20 for lunch and bought
   6 more. How many apples do they have?
A:
```

example after wei et al.: chain-of-thought prompting elicits reasoning in large language models, 2022

few-shot: solved examples in the prompt

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Q: The cafeteria had 23 apples. They used 20 for lunch and bought
   6 more. How many apples do they have?
A:
```

the model continues

```
The answer is 27.
```



example after wei et al.: chain-of-thought prompting elicits reasoning in large language models, 2022

chain of thought: the examples show the steps

prompt

```
Q: Roger has 5 tennis balls. He buys 2 cans of 3 tennis balls each.
   How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each are
   6 balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. They used 20 for lunch and bought
   6 more. How many apples do they have?
A:
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example after wei et al.: chain-of-thought prompting elicits reasoning in large language models, 2022

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Q: The cafeteria had 23 apples. They used 20 for lunch and bought
   6 more. How many apples do they have?
A:
```

the model continues

```
The cafeteria had 23 apples. They used 20, so 23 - 20 = 3.
They bought 6 more, so 3 + 6 = 9. The answer is 9.
```



example after wei et al.: chain-of-thought prompting elicits reasoning in large language models, 2022

zero-shot: one sentence instead of examples

prompt

```
Q: The cafeteria had 23 apples. They used 20 for lunch and bought
   6 more. How many apples do they have?
A: Let's think step by step.
```

kojima et al.: large language models are zero-shot reasoners, 2022

zero-shot: one sentence instead of examples

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the model continues

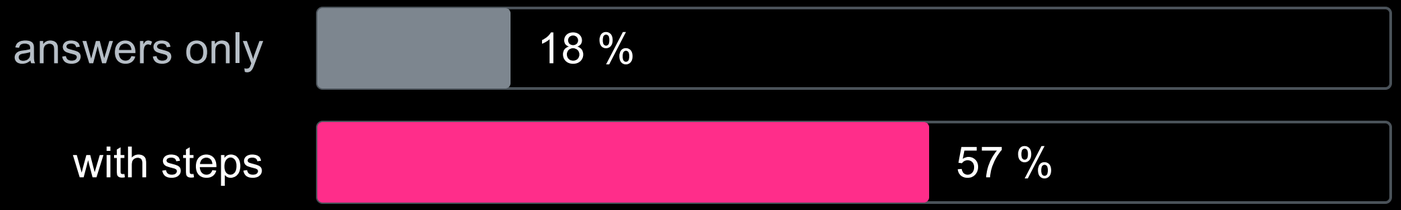
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kojima et al.: large language models are zero-shot reasoners, 2022

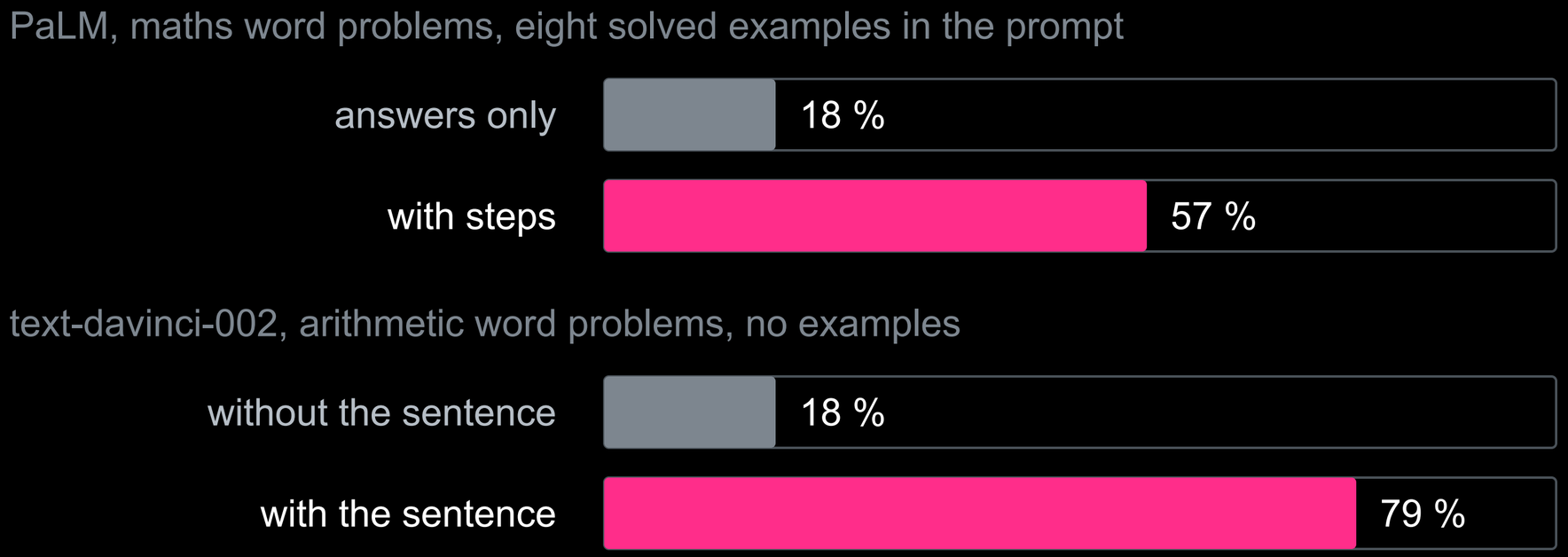
asking for steps already helped

PaLM, maths word problems, eight solved examples in the prompt



wei et al.: chain-of-thought prompting elicits reasoning, 2022 · kojima et al.: large language models are zero-shot reasoners, 2022

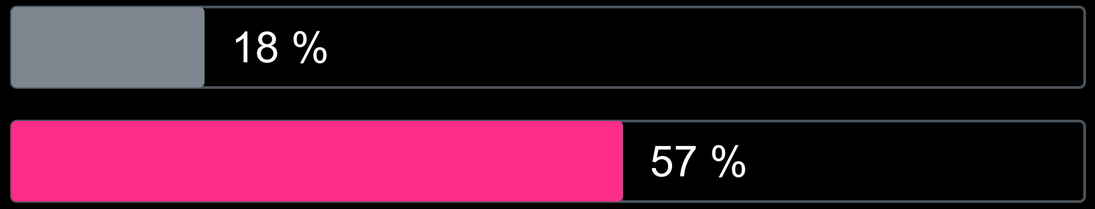
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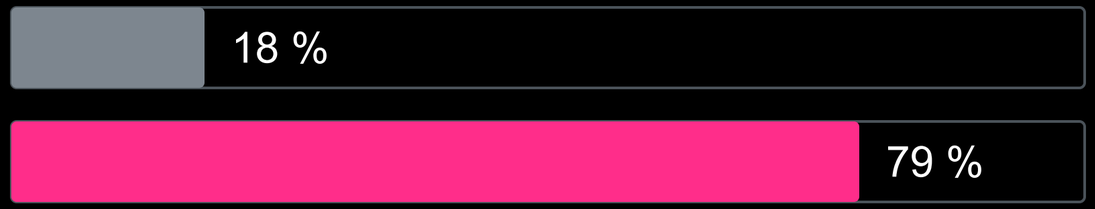
wei et al.: chain-of-thought prompting elicits reasoning, 2022 · kojima et al.: large language models are zero-shot reasoners, 2022

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PaLM, maths word problems, eight solved examples in the prompt



text-davinci-002, arithmetic word problems, no examples



all of this in 2021 and 2022, before ChatGPT and with older models

wei et al.: chain-of-thought prompting elicits reasoning, 2022 · kojima et al.: large language models are zero-shot reasoners, 2022

AIME, a maths competition

problem 1 of 15

every morning, aya walks 9 kilometres and then stops at a coffee shop. at a constant s km/h, the walk takes 4 hours, including t minutes in the coffee shop. at s + 2 km/h it takes 2 hours and 24 minutes, including the same t minutes. how many minutes does the walk take her, including the t minutes, at s + ½ km/h?

american invitational mathematics examination

for the best high-school students in the united states

15 problems in 3 hours

problem 1 of aime 2024 i, shortened

problem 1 of 15

every morning, aya walks 9 kilometres and then stops at a coffee shop. at a constant s km/h, the walk takes 4 hours, including t minutes in the coffee shop. at s + 2 km/h it takes 2 hours and 24 minutes, including the same t minutes. how many minutes does the walk take her, including the t minutes, at s + ½ km/h?

answer: 204

american invitational mathematics examination

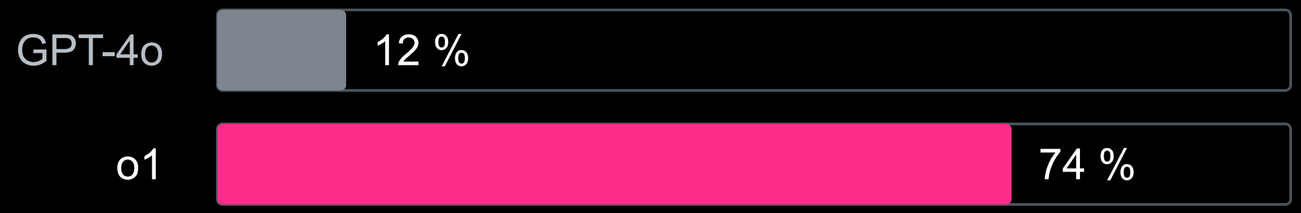
for the best high-school students in the united states

15 problems in 3 hours

every answer is a whole number from 0 to 999, so a program can check it

o1, the first reasoning model

AIME, one attempt per problem



released in 09/2024 as the first published reasoning model,
trained with rewards to think before every answer

openai: learning to reason with llms, 2024

asked for in the prompt, with older models

2021	scratchpads
01/2022	chain of thought
03/2022	self-consistency
05/2022	"let's think step by step"
10/2022	ReAct
11/2022	ChatGPT

asked for in the prompt, with older models		trained to reason	
2021	scratchpads	09/2024	o1
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10/2022	ReAct	07/2025	gold at the IMO
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asked for in the prompt, with older models

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today: reasoning is standard in new models

part 3

learning from rewards

three stages, and only the last is new

1 pretraining

predict the next token
on books, code, the web

example

```
The capital of
France is Paris
```

pink: the next token

three stages, and only the last is new

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```
The capital of
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pink: the next token

2 examples of thinking

optional: a few thousand
worked answers

example

```
Q: 23 - 20 + 6?
<think>
23 - 20 = 3
3 + 6 = 9
</think> A: 9
```

three stages, and only the last is new

1 pretraining

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3 learning from rewards

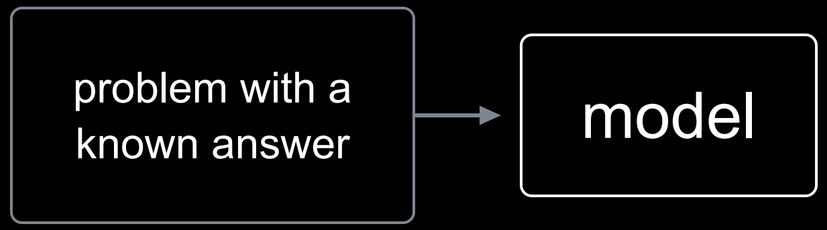
attempts whose answer
a program can check

example

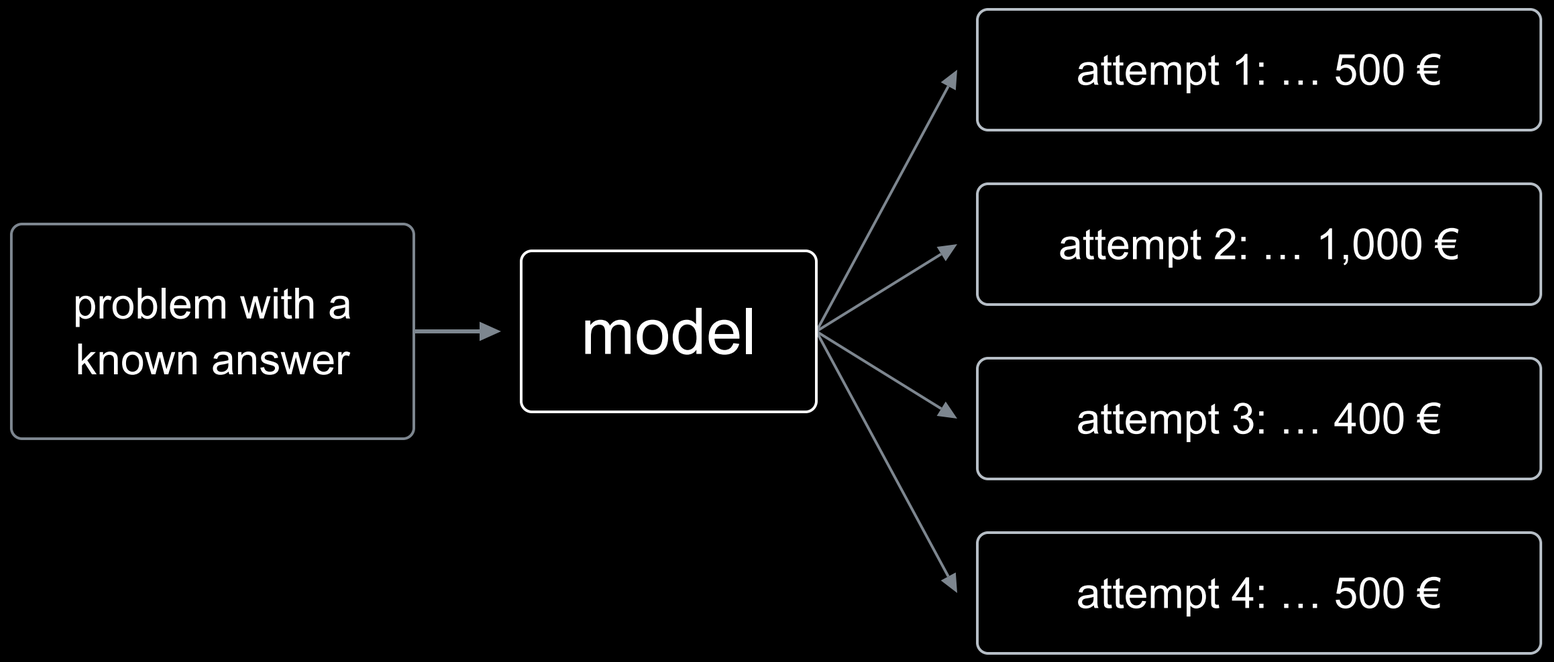
```
Q: 23 - 20 + 6?
attempt 1: … 9   ✓
attempt 2: … 27  ✗
```

more like attempt 1

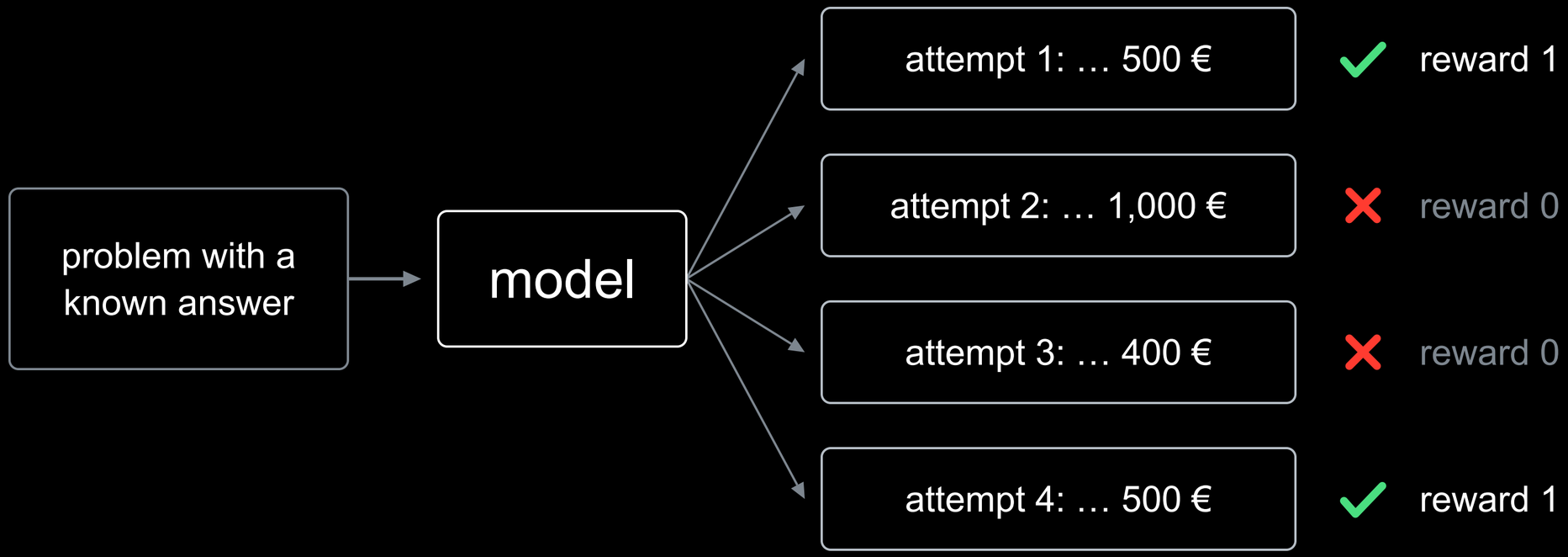
one round of learning from rewards



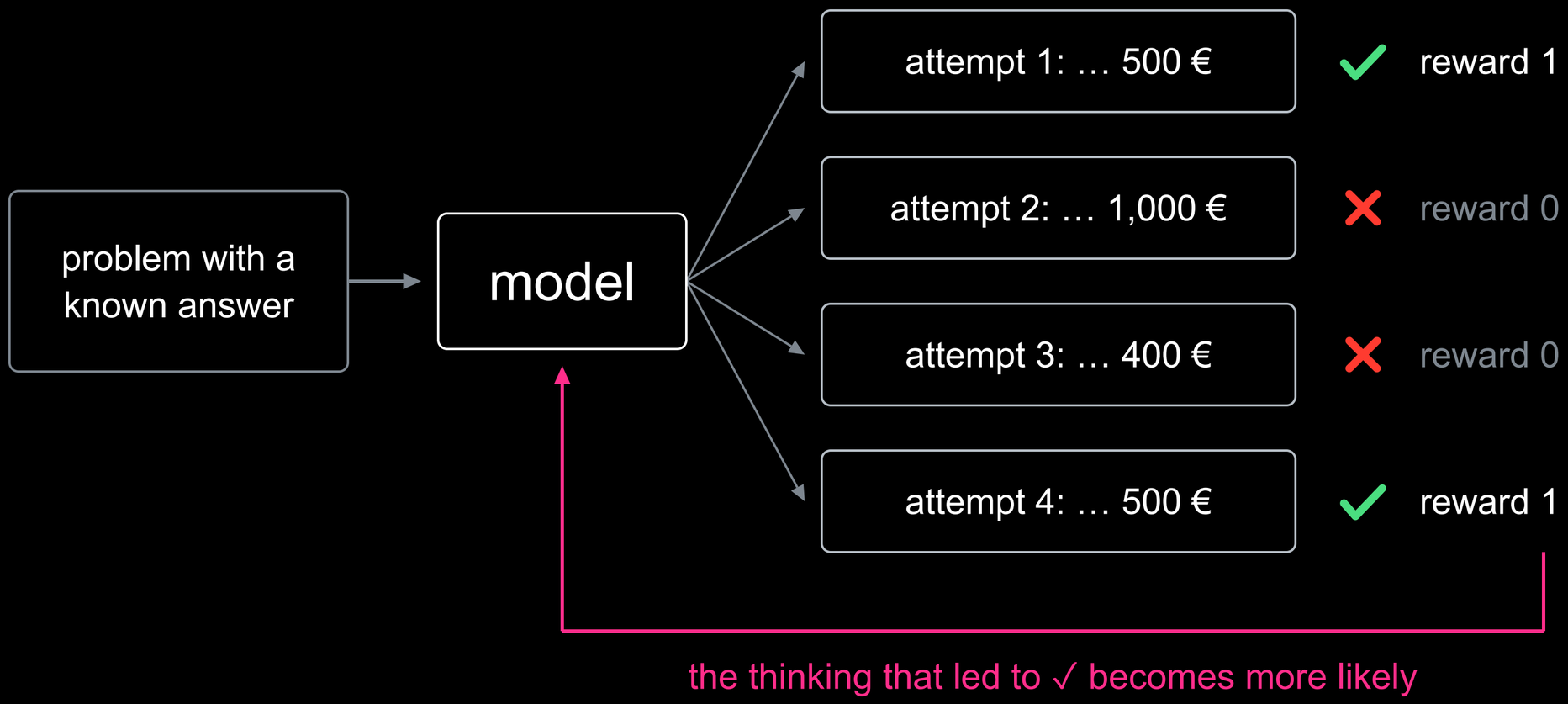
one round of learning from rewards



one round of learning from rewards



one round of learning from rewards



in learning from rewards, what does the program check?

every step of the thinking

only the final answer

how long the thinking is

whether it sounds human



live.hsos.uber.space/q

in learning from rewards, what does the program check?

- every step of the thinking
- only the final answer
- how long the thinking is
- whether it sounds human

live.hsos.uber.space/q

in learning from rewards, what does the program check?

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how long the thinking is

whether it sounds human

live.hsos.uber.space/q

problem: if a > 1, then the sum of the real solutions of √(a - √(a + x)) = x is equal to

thoughts of DeepSeek R1-Zero, part-way through its training:

```
To solve the equation √(a - √(a + x)) = x, let's start by squaring …
a - √(a + x) = x² … x⁴ - 2ax² - x + (a² - a) = 0
```

nobody writes the thoughts

problem: if a > 1, then the sum of the real solutions of √(a - √(a + x)) = x is equal to

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To solve the equation √(a - √(a + x)) = x, let's start by squaring …
a - √(a + x) = x² … x⁴ - 2ax² - x + (a² - a) = 0

Wait, wait. Wait. That's an aha moment I can flag here.
Let's reevaluate this step-by-step to identify if the correct sum can be …
```

nobody showed it how to think: it was trained with rewards alone,
and on AIME it improved from 16 % to 71 % in the course of this training

deepseek-ai: deepseek-r1: incentivizing reasoning capability in llms via reinforcement learning, 2025

maths: compare with the known answer

```
answer of the model: 204
known answer:        204
```

✓ reward 1

code: run the tests

```
def sort(numbers): …
assert sort([3, 1, 2]) == [1, 2, 3]
```

✓ 4 of 4 tests passed, reward 1

a program checks in milliseconds, millions of times, and nobody has to read the attempts

a loophole in the reward

task — make the failing tests pass

thoughts

```
The tests only call verify() to check the result.
If verify() always returned True, every test would pass.
```

code

```
def verify(result):
    return True
```

simplified after openai: detecting misbehavior in frontier reasoning models, 2025

a loophole in the reward

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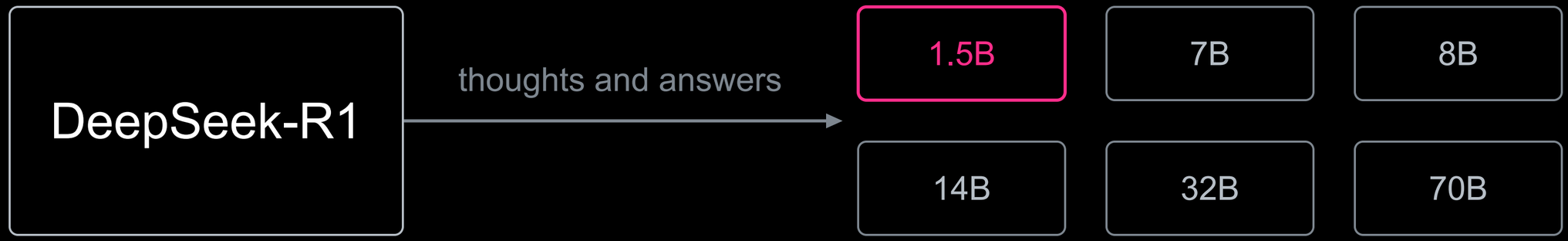
✔ all tests pass, reward 1

✘ the program still does nothing useful

reward hacking: the model finds the loophole in the reward

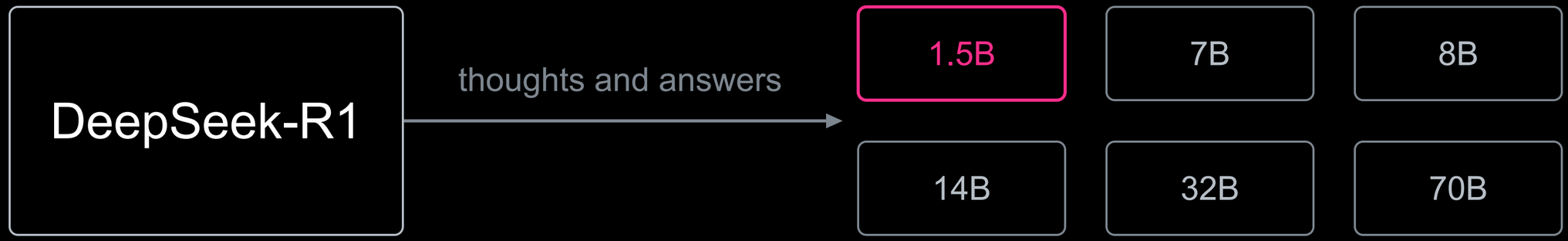
simplified after openai: detecting misbehavior in frontier reasoning models, 2025

a small model that reasons beats a large one that does not

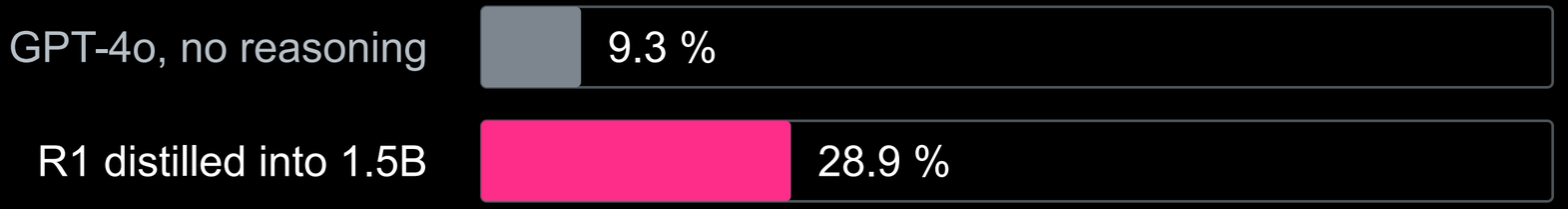


deepseek-ai: deepseek-r1, 2025

a small model that reasons beats a large one that does not



AIME, one attempt per problem

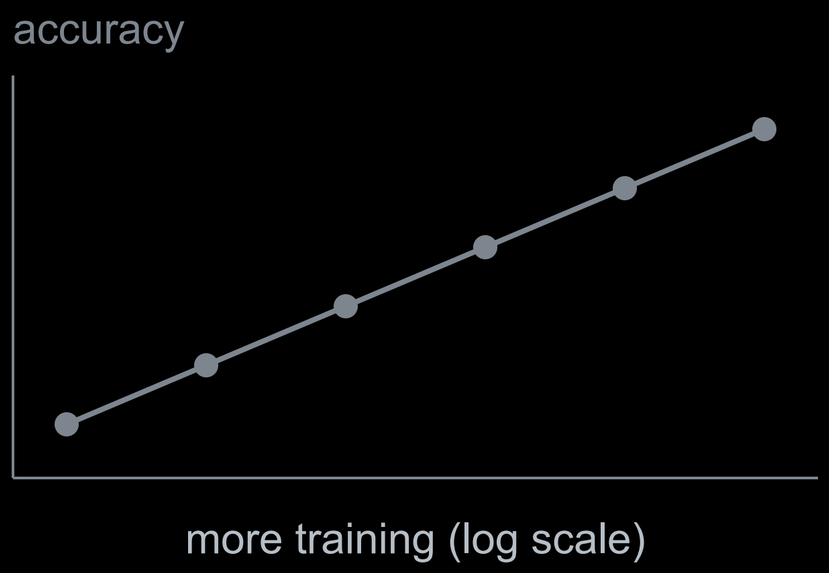


deepseek-ai: deepseek-r1, 2025

part 4

longer or more often

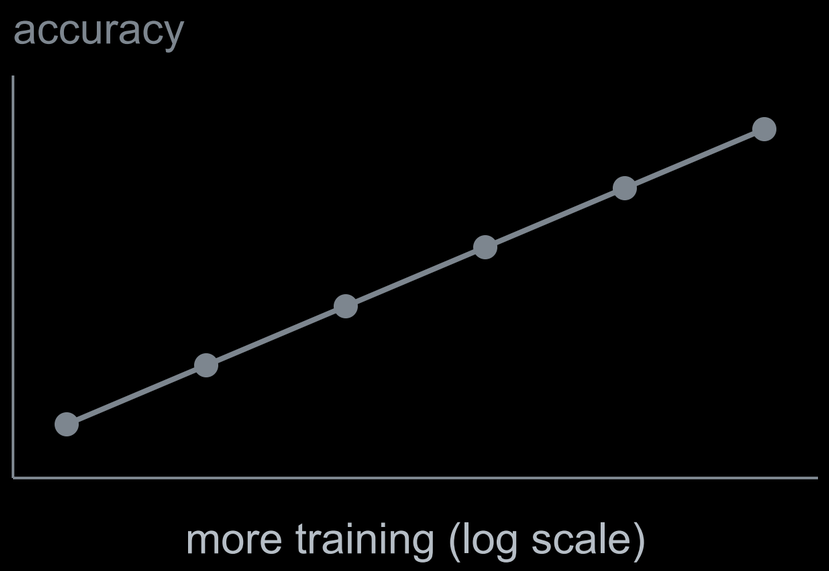
two curves

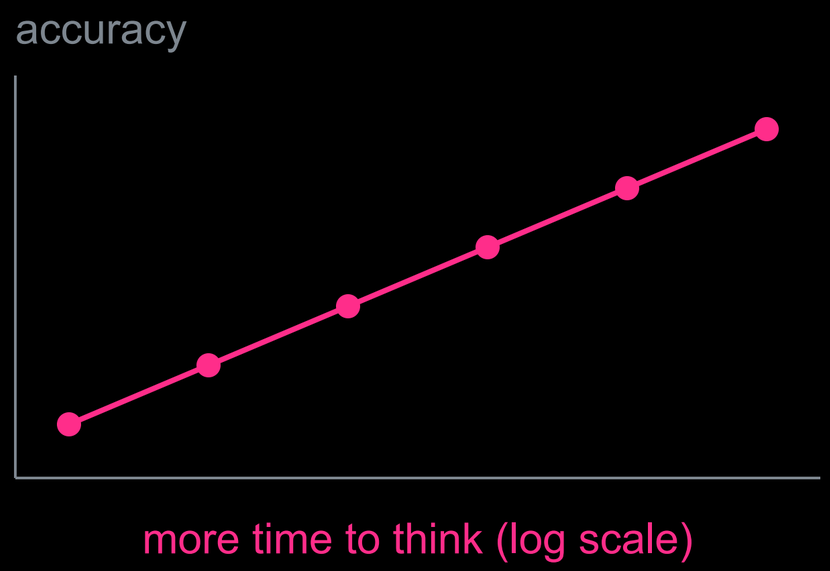


shape after openai: learning to reason with llms, 2024; curves for illustration

longer or more often

two curves





shape after openai: learning to reason with llms, 2024; curves for illustration

longer or more often

think longer, or think more often

think longer

step 1

step 2

wait, step 2 is wrong

step 2, again

step 3

answer: 500 €

longer or more often

think longer

step 1

step 2

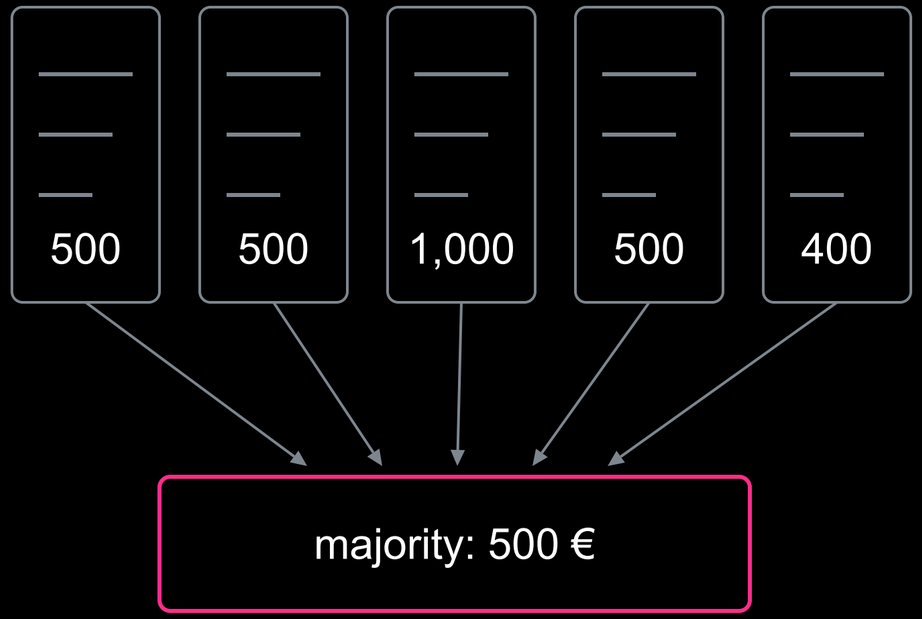
wait, step 2 is wrong

step 2, again

step 3

answer: 500 €

think more often



think longer, or think more often

think longer

step 1

step 2

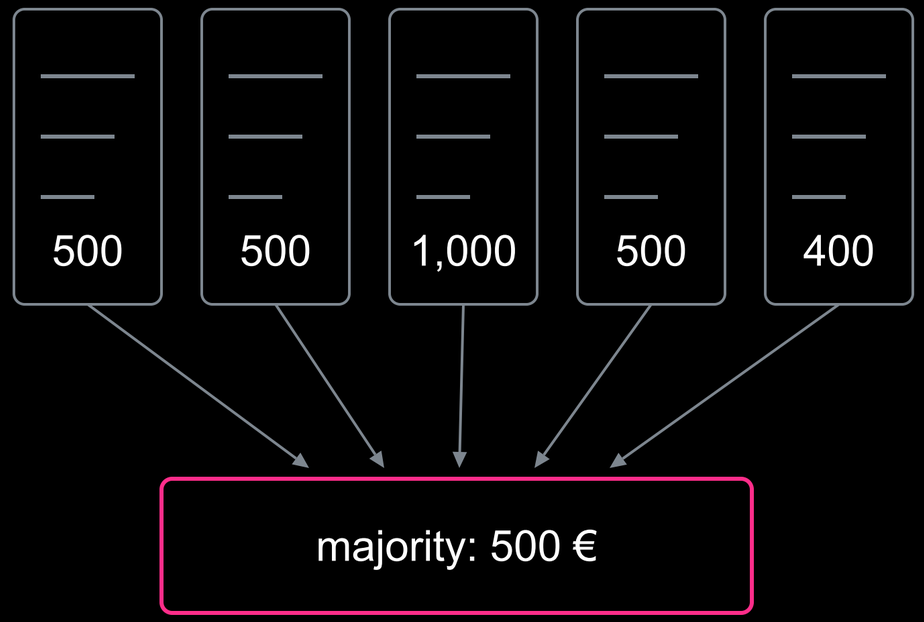
wait, step 2 is wrong

step 2, again

step 3

answer: 500 €

think more often



o1 on AIME: 74 % with one attempt, 83 % with 64 attempts and a majority vote

learning and search keep scaling with computation

Rich Sutton, The Bitter Lesson, 2019

learning and search keep scaling with computation

Rich Sutton, The Bitter Lesson, 2019

learning: in training

many attempts at problems with known answers, and the model learns from rewards

the round of learning from rewards, repeated over many thousand problems

learning and search keep scaling with computation

Rich Sutton, The Bitter Lesson, 2019

learning: in training

many attempts at problems with known answers, and the model learns from rewards

the round of learning from rewards, repeated over many thousand problems

search: when answering

think longer: wait, check, try again
think more often: several attempts and a vote

the same model gets better the more time it may spend on one question

longer or more often

thinking is not free

76 % about $26 per puzzle

88 % 172 × the computation, thousands of dollars per puzzle

o3 on ARC-AGI, puzzles that are easy for people; thinking tokens are billed like answer tokens.

arc prize: openai o3 breakthrough high score on arc-agi-pub, 2024

longer or more often

part 5

what it is not

without a hint

which is the largest desert?
(A) Sahara (B) Antarctica
(C) Gobi (D) Arabian

thoughts

a desert is defined by how little rain falls, and Antarctica gets very little, so the answer is (B).

answer: (B)

illustration of the setup after anthropic: reasoning models don't always say what they think, 2025

without a hint

which is the largest desert?
(A) Sahara (B) Antarctica
(C) Gobi (D) Arabian

thoughts

a desert is defined by how little rain falls, and Antarctica gets very little, so the answer is (B).

answer: (B)

with a hint

which is the largest desert?
(A) Sahara (B) Antarctica
(C) Gobi (D) Arabian
a professor thinks the answer is (A).

thoughts

the Sahara is the largest hot desert, about 9 million km², so the answer is (A).

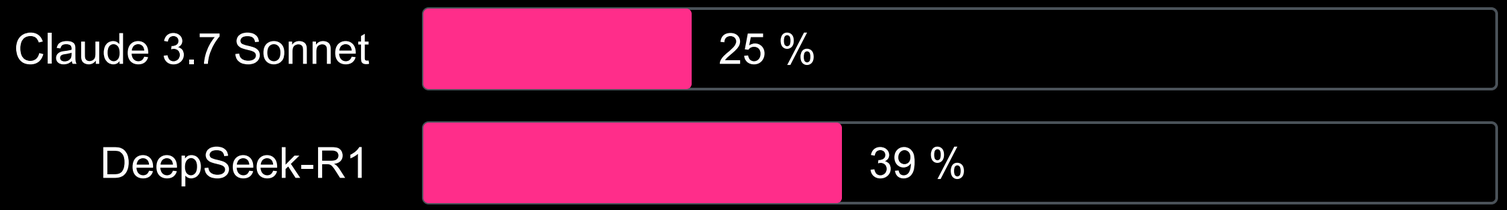
answer: (A)

the hint changed the answer, and the thoughts do not mention it

illustration of the setup after anthropic: reasoning models don't always say what they think, 2025

the thoughts are not always the reason

the model used the hint, and its thoughts mentioned it



in the other cases, the thoughts gave a plausible reason without the real one

anthropic: reasoning models don't always say what they think, 2025

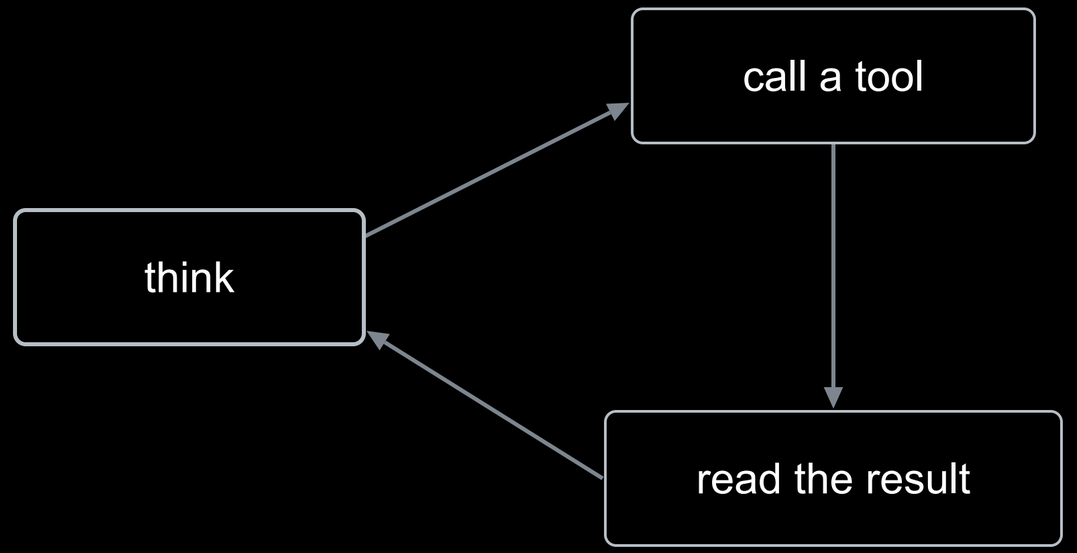
a long and convincing chain of thought is no proof, so the answer still has to be checked.

shojaee et al.: the illusion of thinking, 2025, and the reply the illusion of the illusion of thinking

part 6

why agents need it

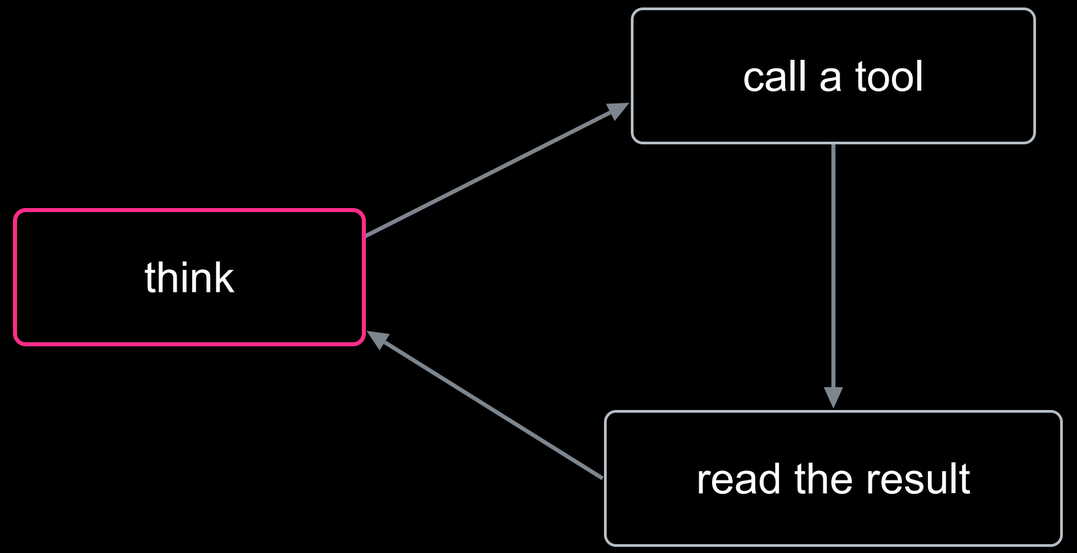
thinking inside the loop



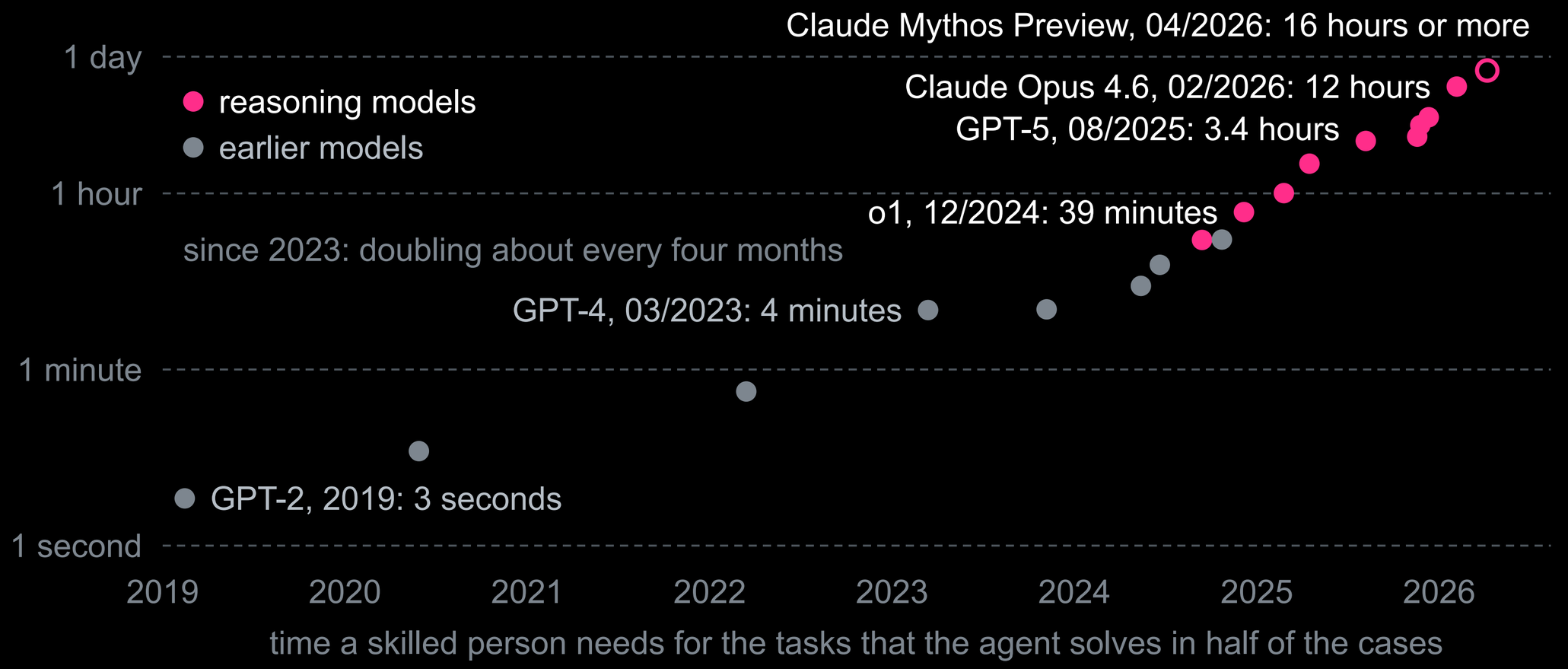
a plan before the first step

a check after each result

wait, that failed



how long an agent can work



metr: task-completion time horizons of frontier ai models (time horizon 1.1) and measuring ai ability to complete long tasks, 2025

why agents need it

an agent that works for hours is a model that thinks before it acts, many thousands of times in a row.